

Integration and Accumulation

The second half of calculus, and the one physics actually spends most of its time doing.

WHERE WE ARE

Chapter 0.1 answered *how fast is this changing?* This chapter answers the question pointing the other way: *given the rate, what accumulated?* You are handed a velocity and asked for a distance; a power and asked for an energy; a concentration and asked for a total exposure.

Those two questions look unrelated. One is about the behaviour of a function in an infinitesimal neighbourhood of a single point; the other is about a total swept out over a whole interval. The central result of this chapter — the Fundamental Theorem of Calculus — says they are the **same question**. That is not a convenience. It is the first appearance of a pattern that recurs at every level of this book, and by Chapter 0.7 it will have grown into Stokes' theorem, which is the single most reused structural fact in theoretical physics.

One practical warning while you're here. Physics is written in differential equations, so it is tempting to think differentiation is the main event. It isn't. **The laws are differential; the answers are integrals.** Newton's second law is a statement about a second derivative, but every actual trajectory is obtained by integrating it twice. Every observable in quantum field theory is an integral. You will differentiate a few hundred times in this book and integrate a few thousand.

TOOLS YOU'LL NEED — Chapter 0.1, specifically the idea that a differentiable function is *locally linear* with an error that dies faster than the displacement. Everything below is that one fact, run in reverse or run in bulk.

1 · Accumulation: the integral as a limit of sums

Start with something you can picture. A car has velocity $v(t)$, which is not constant. How far does it travel between $t = a$ and $t = b$?

If v were constant the answer would be $v \cdot (b - a)$ and there would be no chapter. Since it isn't, do the only honest thing available: chop the trip into N short intervals, each of length $\Delta t = (b - a)/N$. On a short enough interval the velocity *barely changes* — that is exactly the content of continuity — so on the i -th interval the distance covered is close to $v(t_i) \Delta t$ for any t_i inside it. Add them up:

$$\text{distance} \approx \sum_{i=1}^N v(t_i) \Delta t. \quad (0.2.1)$$

Geometrically each term is the area of a rectangle of height $v(t_i)$ and width Δt , so the sum is the area of a staircase drawn under the graph of v . The approximation is bad for small N and improves as the steps get finer, because the amount by which v can wander inside one interval shrinks with the interval.

The definite integral is what (0.2.1) converges to. Written out properly, for a function f on $[a, b]$: choose a **partition** $a = x_0 < x_1 < \dots < x_N = b$, write $\Delta x_i = x_i - x_{i-1}$, define the **mesh** $\|P\| = \max_i \Delta x_i$, and pick a **tag** $x_i^* \in [x_{i-1}, x_i]$ in each cell. Then

$$\int_a^b f(x) \, dx = \lim_{\|P\| \rightarrow 0} \sum_{i=1}^N f(x_i^*) \Delta x_i, \quad (0.2.2)$$

whenever that limit exists.

What "the limit exists" is doing here

As in Chapter 0.1, the escape clause is the interesting part. The demand in (0.2.2) is severe: there must be a single number I such that *every* partition with small enough mesh, tagged *any way you like*, gives a sum within ε of I . You do not get to choose convenient tags. A function that passes this test is called **Riemann integrable**.

Continuous functions on a closed bounded interval pass, and the reason is worth seeing because it is the only place in this chapter where compactness does real work. On $[a, b]$ a continuous function is *uniformly* continuous: for any ε there is a single δ that works everywhere at once. So take any partition of mesh less than δ . Inside cell i the values of f all lie within ε of each other, so the largest possible sum and the smallest possible sum on that cell differ by at most $\varepsilon \Delta x_i$. Adding up,

$$(\text{upper sum}) - (\text{lower sum}) \leq \varepsilon \sum_i \Delta x_i = \varepsilon (b - a). \quad (0.2.3)$$

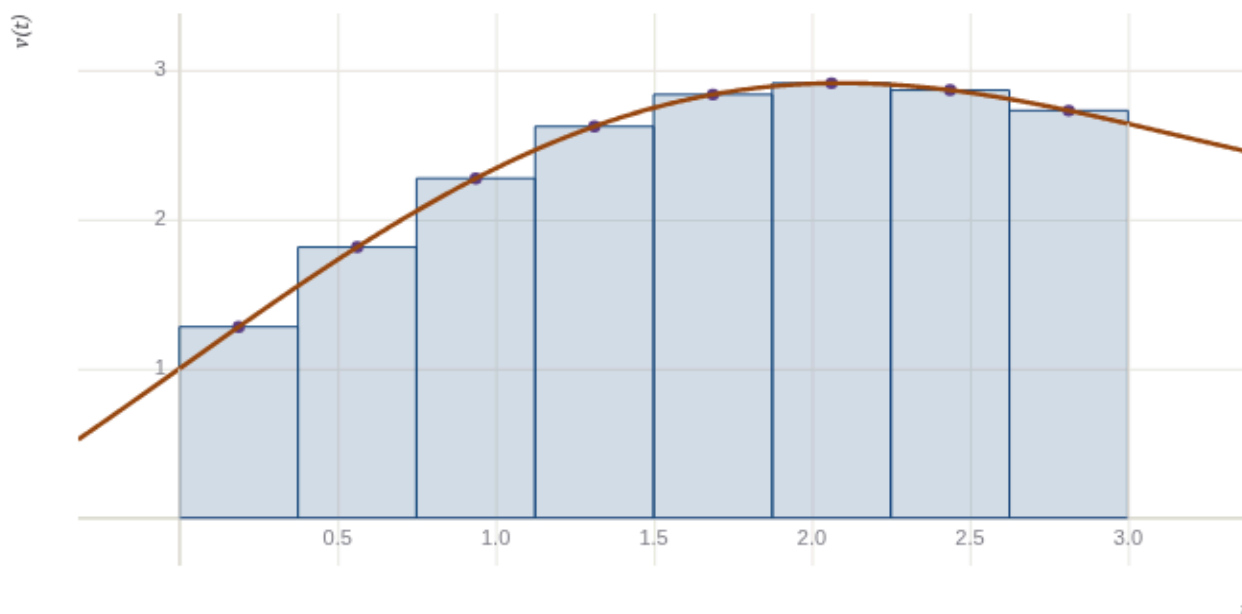
Every tagged sum is trapped between the upper and lower sums, and (0.2.3) squeezes that gap to zero. So all of them converge to the same number. Continuity was not decoration; it is the hypothesis that makes the definition work.

Now a function that fails. Let $\chi(x)$ be 1 when x is rational and 0 when it is irrational. Every interval, however short, contains both kinds of number. So on any partition you can tag entirely at rationals and get $\sum 1 \cdot \Delta x_i = b - a$, or tag entirely at irrationals and get 0. The sums do not converge to anything; χ is not Riemann integrable, at any mesh, ever.

⚠ THIS DEFINITION GETS REPLACED

In Chapter 4.3 we will throw (0.2.2) away and rebuild the integral from scratch (the **Lebesgue** integral), and the reason is visible already. Enumerate the rationals as $q_1, q_2, \dots$ and let f_n be the function equal to 1 at $q_1, \dots, q_n$ and 0 elsewhere. Each f_n is zero except at finitely many points, so each is perfectly Riemann integrable with $\int_0^1 f_n = 0$. But $f_n \rightarrow \chi$ pointwise — and χ is not Riemann integrable at all.

So the Riemann integral is *not closed under limits of functions*. That is fatal the moment you want to treat functions as vectors in a space and ask whether a sequence of them converges — which is precisely what quantum mechanics requires, since a quantum state is a vector in a space of functions. Lebesgue's construction fixes exactly this, and gives $\int_0^1 \chi = 0$ into the bargain. For every function you will meet in the next three parts, the two integrals agree; you need the new one for the architecture, not for the answers.



rectangles N $N = 8$

left

midpoint

right

sum = 7.251701 exact = 7.239992 error = 1.171e-2

$N \times \text{error} = 0.0937$ $N^2 \times \text{error} = 0.7493$

The staircase converging — and the rate at which it converges. The curve is $v(t) = 1 + \frac{1}{2}t + \sin t$ on $[0, 3]$; the shaded staircase is the tagged sum of (0.2.1). Slide N and the sum walks toward the exact value, as (0.2.2) promises. The two readouts on the right are where the physics is. With **left** or **right** tagging, $N \times \text{error}$ settles on a constant (about ∓ 2.462) and $N^2 \times \text{error}$ blows up: the error is $O(1/N)$. Switch to **midpoint** and it inverts — $N \times \text{error}$ crashes to zero while $N^2 \times \text{error}$ settles on about $+0.746$. Midpoint is a whole power of N better for free, and §1.1 explains why in one sentence borrowed from Chapter 0.1.

1.1 · Why midpoint wins, and why you already knew

The left rule approximates f on each cell by a *constant*, $f(x_i)$. Chapter 0.1 tells you the error in a constant approximation is first order in the cell width: f drifts by about $f'(x_i)\Delta x$ across the cell. Each cell therefore contributes an error of order Δx^2 , and there are $N \sim 1/\Delta x$ of them, so the total error is order $\Delta x \sim 1/N$.

The midpoint rule also uses a constant — but it uses the value at the *centre*. And the linear part of f , measured from the centre, is odd: it overshoots on the right of the midpoint by exactly as much as it undershoots on the left. Those two errors cancel *exactly*, not approximately. So the midpoint rule is, silently, as accurate as the best *linear* approximation on each cell, and its leading error is the first thing linearization misses — the quadratic term $\frac{1}{2}f''h^2$ that Chapter 0.1 identified as the leftover. Cell error of order Δx^3 , times N cells, gives $\Delta x^2 \sim 1/N^2$.

This is worth internalising because the move recurs constantly: **put your sample point where the linear error is antisymmetric and it cancels**. It is the same reason odd moments of a Gaussian vanish in §4, the same reason a symmetric difference quotient is second-order accurate, and the same reason the leading correction in a saddle-point expansion (Chapter 5.5) is quadratic rather than linear.

▼ Grind box — the error constants, exactly, so you can check them against the figure

Work on one cell of width d and expand f using Chapter 0.1's quadratic form.

Left rule. With the cell $[x, x + d]$, the exact contribution is

$$\int_0^d f(x + s) ds = \int_0^d \left[f(x) + f'(x)s + O(s^2) \right] ds = f(x)d + \frac{1}{2}f'(x)d^2 + O(d^3),$$

while the estimate is $f(x)d$. So the per-cell error (estimate minus truth) is $-\frac{1}{2}f'(x)d^2$. Sum over cells and recognise the leftover sum as another Riemann sum:

$$E_{\text{left}} \approx -\frac{1}{2}d^2 \sum_i f'(x_i) = -\frac{d}{2} \sum_i f'(x_i) d \rightarrow -\frac{d}{2} \int_a^b f' = -\frac{(b-a)[f(b) - f(a)]}{2N}.$$

For the figure's f on $[0, 3]$: $f(3) - f(0) = 1.64112$, so $N \times E_{\text{left}} \rightarrow -3(1.64112)/2 = -2.4617$. That is the number in the readout. The right rule is the same computation with the opposite sign, $+2.4617$.

Midpoint. Expand about the centre m and integrate symmetrically. The odd term integrates to zero:

$$\int_{-d/2}^{d/2} \left[f(m) + f'(m)s + \frac{1}{2}f''(m)s^2 \right] ds = f(m)d + 0 + \frac{f''(m)d^3}{24},$$

using $\int_{-d/2}^{d/2} s ds = 0$ and $\int_{-d/2}^{d/2} s^2 ds = d^3/12$. The estimate is $f(m)d$, so the per-cell error is $-f''(m)d^3/24$ and

$$E_{\text{mid}} \rightarrow -\frac{d^2}{24} \int_a^b f'' = -\frac{(b-a)^2 [f'(b) - f'(a)]}{24N^2}.$$

Here $f'(t) = \frac{1}{2} + \cos t$, so $f'(3) - f'(0) = -1.98999$ and $N^2 \times E_{\text{mid}} \rightarrow -9(-1.98999)/24 = +0.7462$. Also in the readout.

Trapezoid, and a bonus. Averaging the two endpoint values gives, after expanding both about m , an estimate $f(m)d + f''(m)d^3/8$, hence a per-cell error of $+f''(m)d^3/12$ — *exactly -2 times the midpoint error*. So the combination

$$S = \frac{1}{3}(2M + T)$$

annihilates the d^3 term identically. That is Simpson's rule, and it is not a lucky guess: it is the unique weighting that kills the leading error of two rules whose errors are in a known fixed ratio. The same manoeuvre — combining two approximations to cancel a known leading error — is called Richardson extrapolation, and you will meet its physics incarnation in Chapter 5.9, where the "known leading error" is a divergence and cancelling it is renormalization.

2 · The Fundamental Theorem, both halves, derived

Nothing so far connects (0.2.2) to derivatives. The definition is about areas and sums; Chapter 0.1 was about local slopes. Watching those two collapse into one statement is the point of this section.

2.1 · First half: differentiating an accumulation

Let f be continuous on $[a, b]$ and define the **accumulation function**

$$F(x) = \int_a^x f(t) dt. \quad (0.2.4)$$

$F(x)$ is the total accumulated between the fixed left end a and the moving right end x . Ask Chapter 0.1's question about it: how fast does F change as you slide x ?

Push x along by h . The accumulation from a to $x + h$ is the accumulation from a to x plus whatever the sliver from x to $x + h$ contributes — that is just additivity of the sum in (0.2.2) over adjacent blocks of cells:

$$F(x + h) - F(x) = \int_x^{x+h} f(t) dt. \quad (0.2.5)$$

Now estimate the sliver. Because f is continuous at x , write $f(t) = f(x) + \eta(t)$ where $\eta(t) = f(t) - f(x)$, and let

$$\varepsilon(h) = \max_{|t-x| \leq |h|} |f(t) - f(x)| \quad \text{so that} \quad \varepsilon(h) \rightarrow 0 \text{ as } h \rightarrow 0. \quad (0.2.6)$$

That last statement is the definition of continuity at x , nothing more. Split the sliver:

$$\int_x^{x+h} f = \underbrace{f(x)h}_{\text{rectangle}} + \underbrace{\int_x^{x+h} \eta(t) dt}_{\text{leftover}}, \quad \left| \int_x^{x+h} \eta \right| \leq \varepsilon(h) |h|. \quad (0.2.7)$$

The bound on the leftover follows straight from (0.2.2): every tagged sum for $\int \eta$ is at most $\varepsilon(h)$ times the total width $|h|$, so the limit is too. Combining (0.2.5) and (0.2.7):

$$F(x+h) = F(x) + \underbrace{f(x)h}_{\text{smaller than } h} + \underbrace{h\varepsilon(h)}_{\text{smaller than } h}. \quad (0.2.8)$$

Look at what that is. It is *literally* Chapter 0.1's boxed equation — value, plus something linear in h , plus an error that dies faster than h — with the coefficient of h equal to $f(x)$. And Chapter 0.1 established that the coefficient with that property is unique and is called the derivative. So, reading it off:

$$\boxed{\frac{d}{dx} \int_a^x f(t) dt = f(x).} \quad (0.2.9)$$

No trickery, no area formulas, no pictures required. The sliver is a thin rectangle of height $f(x)$ and width h ; the error you make by pretending it is *exactly* a rectangle is second order, so it dies. Same move as the corner square in the product rule, same move as everything in Chapter 0.1.

2.2 · Second half: evaluating an integral

(0.2.9) is a statement about a function you build by integrating. The version you actually compute with runs the other way. Suppose you can find *any* function G with $G' = f$ — call it an **antiderivative**. Then G and the F of (0.2.4) have the same derivative, so

$$(G - F)' = f - f = 0 \quad \implies \quad G(x) - F(x) = C \quad (\text{a constant}). \quad (0.2.10)$$

(That a function with vanishing derivative on an interval must be constant is not free; it comes from the mean value theorem, which is derived in the grind box below.) Now evaluate at both ends and subtract, so that the unknown constant cancels:

$$G(b) - G(a) = [F(b) + C] - [F(a) + C] = F(b) - F(a) = \int_a^b f - 0, \quad (0.2.11)$$

using $F(a) = \int_a^a f = 0$. So

$$\boxed{\int_a^b f(x) dx = G(b) - G(a) \equiv \left[G(x) \right]_a^b, \quad \text{for any } G \text{ with } G' = f.} \quad (0.2.12)$$

This is the reason you can compute integrals at all. The limit of sums in (0.2.2) is an infinite process with no obvious shortcut; (0.2.12) replaces it with two evaluations of a function you found by running the differentiation rules backwards. Every integral you have ever computed by hand was this theorem being spent.

THE PUNCHLINE, AND IT IS NOT A SMALL ONE

Here is (0.2.12) read differently. Take a partition of $[a, b]$ and telescope:

$$G(b) - G(a) = \sum_{i=1}^N [G(x_i) - G(x_{i-1})] \approx \sum_{i=1}^N G'(x_{i-1}) \Delta x_i \longrightarrow \int_a^b G'.$$

Every interior value $G(x_i)$ appears twice with opposite signs and cancels. Only the two *endpoints* survive. That is the entire mechanism: **you add up local changes, the interior cancels in pairs, and what is left lives on the boundary.**

So the Fundamental Theorem says a *local* quantity (the derivative, defined at each point) integrated over a *region* equals a *global* quantity evaluated only on the *boundary* of that region. In the notation of Chapter 0.7 this reads

$$\int_{[a,b]} dG = \int_{\partial[a,b]} G, \quad \partial[a,b] = \{b\} - \{a\},$$

and that equation, unchanged in form, is **Stokes' theorem**. Raise the dimension and the same cancellation of interior contributions gives Green's theorem, the divergence theorem, and the curl theorem — Chapter 0.7 proves they are one theorem wearing three costumes, and the costume you are wearing right now is the one-dimensional original. In Chapter 3.5 this is how a *local* conservation law $\nabla_\mu T^{\mu\nu} = 0$ gets converted into a statement about total energy in a region, and where that conversion fails in general relativity you get one of the deepest open problems in the subject. All of it is interior terms cancelling in pairs.

▼ Grind box — the fine print: mean value theorem, and where $F' = f$ fails

Zero derivative implies constant. First Rolle's theorem: if g is continuous on $[a, b]$, differentiable inside, and $g(a) = g(b)$, then $g'(c) = 0$ somewhere inside. Why: a continuous function on a closed bounded interval attains a maximum and a minimum. If both occur at the endpoints then g is constant and any interior c works. Otherwise there is an interior extremum at c ; there the difference quotient is ≥ 0 approaching from one side and ≤ 0 from the other, and since the derivative exists both one-sided limits equal it, forcing $g'(c) = 0$.

Now the mean value theorem. Given G on $[a, b]$, apply Rolle to

$$g(x) = G(x) - G(a) - \frac{G(b) - G(a)}{b - a} (x - a),$$

which satisfies $g(a) = g(b) = 0$. The conclusion $g'(c) = 0$ reads $G(b) - G(a) = G'(c)(b - a)$. If $G' \equiv 0$ then every such difference vanishes, so G takes the same value everywhere: constant. That is the step used in (0.2.10).

Where the first half breaks. (0.2.9) used continuity of f at the point x . Drop it and the theorem drops with it. Take $f(t) = \text{sign}(t)$ and $a = -1$. Then $F(x) = |x| - 1$, which is perfectly continuous everywhere but has no derivative at 0 — precisely Chapter 0.1's corner. The accumulation function exists; it just isn't differentiable where the integrand jumps.

Notice the direction of that asymmetry, because it matters later. Differentiating makes a function *rougher*: a corner in F becomes a jump in f . Integrating makes it *smoother*: a jump in f becomes a mere corner in F . This is why integral formulations of physical laws are better behaved than differential ones, why "weak solutions" of PDEs are defined by integrating against test functions, and why the Dirac delta of Chapter 0.9 — an object with no pointwise meaning at all — is perfectly well defined the moment it appears inside an integral.

Fine print on the second half. (0.2.12) as derived assumed f continuous, so that F exists and $F' = f$. A slightly stronger version holds: if f is merely Riemann integrable and happens to have *some* antiderivative G , then $\int_a^b f = G(b) - G(a)$ still, by applying the mean value theorem to G on each cell of a partition and recognising the result as a tagged Riemann sum. You will not need the stronger version, but you should know the weaker one has hypotheses.

3 · Techniques, derived as reversed rules

Because of (0.2.12), integrating means recognising something as a derivative. Every differentiation rule from Chapter 0.1 therefore becomes an integration technique when read backwards. There are only two that matter, and one of them matters enormously.

3.1 · Substitution is the chain rule, reversed

Let $F' = f$ and let g be differentiable. The chain rule says

$$\frac{d}{dx} F(g(x)) = f(g(x)) g'(x). \quad (0.2.13)$$

So the right-hand side of (0.2.13) has a known antiderivative, namely $F \circ g$. Feed that into (0.2.12):

$$\int_a^b f(g(x)) g'(x) dx = F(g(b)) - F(g(a)) = \int_{g(a)}^{g(b)} f(u) du. \quad (0.2.14)$$

That is substitution, derived rather than asserted. Two things to notice. First, the informal ritual "let $u = g(x)$, so $du = g'(x)dx$ " is a compressed way of writing exactly (0.2.14) — and the licence to manipulate du that way is the same licence Chapter 0.1 discussed under "one deliberate abuse". Second, **the limits transform too**. Forgetting that is the most common error in the entire technique, and it is why $\int_{-1}^1 x e^{-x^2} dx$ is zero: under $u = x^2$ both limits become 1, and any integral whose two limits coincide is zero. The symmetry argument and the substitution argument agree, as they must.

3.2 · Integration by parts is the product rule, reversed

Take the product rule $(uv)' = u'v + uv'$, integrate both sides from a to b , and apply (0.2.12) to the left (whose integrand is manifestly a derivative):

$$\left[uv\right]_a^b = \int_a^b u'v \, dx + \int_a^b uv' \, dx. \quad (0.2.15)$$

Rearranged, that is **integration by parts**:

$$\boxed{\int_a^b u v' \, dx = \left[uv\right]_a^b - \int_a^b u' v \, dx.} \quad (0.2.16)$$

Read (0.2.16) as an instruction rather than a formula: **you may move a derivative from one factor onto the other, at the cost of a minus sign and a boundary term**. That sentence is, without much exaggeration, the most-used manipulation in theoretical physics. It is not primarily a trick for computing $\int x e^x$. It is the mechanism by which:

- **An action becomes a field equation.** In Chapter 1.2 you vary $S = \int L(q, \dot{q}) dt$ and get a term proportional to $\delta \dot{q}$ — a variation of a *derivative*, which you cannot set to zero independently. By parts moves the time derivative off $\delta \dot{q}$ and onto the coefficient, leaving everything multiplying δq alone. That single step is what turns "the action is stationary" into the Euler–Lagrange equation, and in Chapter 5.2 the identical step in four dimensions turns a field Lagrangian into a field equation. Every field equation in this book is produced by (0.2.16).
- **An operator gets an adjoint.** In Chapter 4.3, $\int \psi^* \partial_x \phi \, dx = - \int (\partial_x \psi)^* \phi \, dx$ when boundary terms vanish. The minus sign is why $\hat{p} = -i\hbar \partial_x$ is Hermitian and ∂_x alone is not — the factor of i exists to absorb the minus sign that (0.2.16) generates. That is the entire origin of the i in the momentum operator.
- **A derivative gets defined for objects that have none.** The derivative of a distribution in Chapter 0.9 is *defined* by declaring (0.2.16) to hold.

▼ Grind box — the bookkeeping, and why "the boundary term vanishes" is physics, not maths

Worked mechanics. $\int_0^\infty x e^{-x} dx$. Choose $u = x$ and $v' = e^{-x}$, so $u' = 1$ and $v = -e^{-x}$. Then

$$\int_0^\infty x e^{-x} dx = \left[-x e^{-x} \right]_0^\infty + \int_0^\infty e^{-x} dx = 0 + 1 = 1.$$

The boundary term vanished at ∞ because e^{-x} beats x , and at 0 because x beats e^{-x} — that is a *check*, performed, not an assumption. Choosing u and v' the other way round gives $\int x e^{-x} = \left[\frac{1}{2} x^2 e^{-x} \right] +$

$\frac{1}{2} \int x^2 e^{-x}$, which is true and useless: the remaining integral is worse than the one you started with. The craft in by parts is entirely in choosing which factor gets differentiated — pick the one that gets *simpler*.

The move that builds physics. Here is the Chapter 1.2 step, done now so it is not new later. Let $S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt$ and perturb the path $q \rightarrow q + \delta q$. To first order,

$$\delta S = \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right] dt.$$

The second term is stuck: $\delta \dot{q}$ is not independent of δq , it is its derivative. Use (0.2.16) with $u = \partial L / \partial \dot{q}$ and $v' = \delta \dot{q}$, noting $\delta \dot{q} = \frac{d}{dt} \delta q$:

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}} \frac{d}{dt} (\delta q) dt = \left[\frac{\partial L}{\partial \dot{q}} \delta q \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \delta q dt.$$

The boundary term dies because we only compare paths with the *same endpoints*, so $\delta q(t_1) = \delta q(t_2) = 0$. What is left is

$$\delta S = \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \right] \delta q dt,$$

and now δq is arbitrary and unconstrained, so demanding $\delta S = 0$ for all of them forces the bracket to vanish pointwise. That bracket is the Euler–Lagrange equation. Notice that *by parts did all the work*: it converted an unusable variation of a derivative into a usable variation of the coordinate.

The honest warning. In field theory you will constantly see $\int \partial_\mu \phi \partial^\mu \phi \rightarrow - \int \phi \square \phi$ with the boundary term "dropped". That is not a mathematical triviality — it is the *physical assumption that the fields fall off fast enough at infinity*. Sometimes they don't, and then the boundary term is the interesting object rather than the discarded one:

- In general relativity the Einstein–Hilbert action needs an explicit boundary term (the Gibbons–Hawking term) added by hand, precisely because the naive one does not vanish (Chapter 3.6).
- Boundary terms at a black hole horizon are where the entropy lives (Chapter 7.8).
- Topological terms in gauge theory are *total derivatives* — pure boundary — and they still change the physics, which is how the strong CP problem exists at all (Chapter 6.5).

So: drop boundary terms freely, but know each time that you have made a physical claim about what happens far away, and be ready to be asked to defend it.

4 · The Gaussian integral

We now derive the single most-used integral in this book. It appears in the normalisation of every quantum wave packet (Chapter 4.4), in the free-particle propagator (Chapter 5.8), and — this is not an overstatement — the entire perturbative expansion of quantum field theory consists of Gaussian integrals plus corrections (Chapter 5.5). Learn it once, properly.

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx. \quad (0.2.17)$$

First, does it even converge? For $|x| \geq 1$ we have $x^2 \geq |x|$, so $e^{-x^2} \leq e^{-|x|}$, and $\int_1^{\infty} e^{-x} dx = e^{-1}$ is finite; on $[-1, 1]$ the integrand is bounded by 1. So I is finite and positive. Good — we are entitled to look for its value.

And we cannot find it the usual way, because e^{-x^2} has no elementary antiderivative (see the callout after §5 — this is a theorem, not a failure of effort). So we do something that looks like cheating and isn't: **square it**.

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2} dy \right) = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy. \quad (0.2.18)$$

Renaming the second dummy variable y is free; combining the two one-dimensional integrals into one two-dimensional integral is the real step, and it is legal because the integrand is positive (grind box). Now the point: $x^2 + y^2$ is the squared distance from the origin. The integrand depends only on distance, so it is begging to be written in polar coordinates.

4.1 · The area element, derived

Set $x = r \cos \theta$, $y = r \sin \theta$, so $x^2 + y^2 = r^2$. The one thing that needs care is what $dx dy$ becomes. A cell of the polar grid runs from r to $r + dr$ and from θ to $\theta + d\theta$. Its area is the difference of two circular sectors, and a sector of radius ρ and angle $d\theta$ has area $\frac{1}{2}\rho^2 d\theta$ (that is what radian measure means):

$$dA = \frac{1}{2}(r + dr)^2 d\theta - \frac{1}{2}r^2 d\theta = r dr d\theta + \underbrace{\frac{1}{2}(dr)^2 d\theta}_{\text{second order}}. \quad (0.2.19)$$

That is exact, and the leftover in (0.2.19) is second order in the cell size, so it dies in the limit — the same corner-square argument as the product rule in Chapter 0.1, wearing a different hat. Hence $dx dy = r dr d\theta$. (Chapter 0.6 will produce the same factor r mechanically, as the Jacobian determinant; it is worth having seen it geometrically first, because the Jacobian is nothing but "how much a coordinate change stretches area", and here you can watch it stretch.)

4.2 · The value

Now the whole plane is covered by $r \in [0, \infty)$, $\theta \in [0, 2\pi)$, and

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta = 2\pi \int_0^{\infty} r e^{-r^2} dr. \quad (0.2.20)$$

And *that* integral is elementary, because the stray factor of r is exactly the g' that substitution wants: with $u = r^2$, $du = 2r \, dr$,

$$\int_0^\infty r e^{-r^2} dr = \frac{1}{2} \int_0^\infty e^{-u} du = \frac{1}{2} \left[-e^{-u} \right]_0^\infty = \frac{1}{2}. \quad (0.2.21)$$

So $I^2 = 2\pi \cdot \frac{1}{2} = \pi$, and since $I > 0$,

$$\boxed{\int_{-\infty}^\infty e^{-x^2} dx = \sqrt{\pi}.} \quad (0.2.22)$$

Pause on why this worked, because the mechanism is more useful than the answer. In one dimension e^{-x^2} has no elementary antiderivative. In two dimensions, after going polar, the integrand acquires a factor of r from the area element — and $r e^{-r^2}$ *does* have one. The trick is not a trick about exponentials; it is the observation that **the geometry of the plane supplies the missing factor**. Nothing in the one-dimensional problem could have told you that π was going to appear. It came from the angular integral, i.e. from the fact that a circle exists.

▼ Grind box — is squaring the integral legal, and the polar limit

Two things were waved at in (0.2.18) and both are fine, but you should know why.

Turning a product of integrals into a double integral. This is Fubini's theorem, and for a non-negative integrand it holds with no extra hypotheses at all (that version is Tonelli's theorem). Our integrand $e^{-(x^2+y^2)}$ is positive everywhere, so we are safe. For *oscillatory* integrands the same step can fail spectacularly, and in Chapter 5.8 the Gaussian we need is e^{iax^2} — purely oscillatory, not decaying. The standard repair is to give a a small imaginary part, do the honest convergent integral, and take the limit at the end. That deformation has a name — the $i\epsilon$ prescription — and it is not bookkeeping: it is what selects the causal propagator over the anticausal one.

The infinite region. Strictly, $\iint_{\mathbb{R}^2}$ means a limit of integrals over expanding bounded regions, and one should check the answer does not depend on *how* you expand. Squares of side $2L$ give I_L^2 where $I_L = \int_{-L}^L e^{-x^2} dx$; discs of radius R give $\pi(1 - e^{-R^2})$ by the same polar computation. Since every disc of radius L sits inside the square of side $2L$, which sits inside the disc of radius $L\sqrt{2}$,

$$\pi(1 - e^{-L^2}) \leq I_L^2 \leq \pi(1 - e^{-2L^2}),$$

and both ends go to π . Squeezed. The answer is independent of the exhaustion, which for a positive integrand it always is.

4.3 • The general Gaussian

Rescale. For $a > 0$ put $u = \sqrt{a} x$, so $du = \sqrt{a} dx$ and the limits are unchanged:

$$\int_{-\infty}^{\infty} e^{-ax^2} dx = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\frac{\pi}{a}}. \quad (0.2.23)$$

Now add a linear term, which is where it starts being useful. The move is to complete the square:

$$-ax^2 + bx = -a \left(x^2 - \frac{b}{a}x \right) = -a \left(x - \frac{b}{2a} \right)^2 + \frac{b^2}{4a}. \quad (0.2.24)$$

The last term is a constant as far as x is concerned, so it comes out of the integral; the first is a shifted Gaussian, and shifting the integration variable over the whole real line changes nothing. Hence

$$\boxed{\int_{-\infty}^{\infty} e^{-ax^2+bx} dx = \sqrt{\frac{\pi}{a}} e^{b^2/4a}.} \quad (0.2.25)$$

Commit (0.2.25) to memory. In Chapter 5.5 the variable x becomes an entire field configuration, a becomes a differential operator, b becomes an external source J , and the formula becomes — with almost no change of shape — $Z[J] = Z[0] \exp \left(\frac{1}{2} \iint J \Delta J \right)$, the generating functional of a free quantum field theory. Δ is the propagator, and it sits exactly where $1/4a$ sits here.

4.4 · Differentiating under the integral sign

What about $\int x^2 e^{-ax^2} dx$? You could hunt for a substitution. Don't. Notice instead that a appears in (0.2.23) as a *parameter*, and that differentiating the integrand with respect to it pulls down exactly the factor you want:

$$\frac{\partial}{\partial a} e^{-ax^2} = -x^2 e^{-ax^2}. \quad (0.2.26)$$

So if we may swap the order of ∂_a and $\int dx$ — and we may, see below — then differentiating both sides of (0.2.23) with respect to a gives an integral we never had to solve:

$$\begin{aligned} - \int_{-\infty}^{\infty} x^2 e^{-ax^2} dx &= \frac{d}{da} \sqrt{\frac{\pi}{a}} = -\frac{1}{2} \sqrt{\pi} a^{-3/2}, \\ \implies \int_{-\infty}^{\infty} x^2 e^{-ax^2} dx &= \frac{1}{2a} \sqrt{\frac{\pi}{a}}. \end{aligned} \quad (0.2.27)$$

Repeat and you get every even moment for free:

$$\int_{-\infty}^{\infty} x^{2n} e^{-ax^2} dx = (-1)^n \frac{d^n}{da^n} \sqrt{\frac{\pi}{a}} = \frac{(2n)!}{n! (4a)^n} \sqrt{\frac{\pi}{a}}, \quad (0.2.28)$$

while every *odd* moment vanishes because $x^{2n+1} e^{-ax^2}$ is an odd function integrated over a symmetric interval — the same cancellation that made the midpoint rule good in §1.1.

This technique deserves a name and a place in your hands: parameter differentiation (sometimes "Feynman's trick", though it long predates him). You introduce a parameter the original problem did not have, differentiate or integrate with respect to it, and trade a hard integral for an easy one plus some calculus. Problem 2 below is the cleanest possible example.

4.5 · The same number, two ways — and why the second way is quantum field theory

There is a second route to (0.2.27) that looks like a detour and is actually the whole of Chapter 5.5. Instead of differentiating with respect to a , differentiate (0.2.25) with respect to the *source* b . Since $\partial_b e^{bx} = x e^{bx}$, each derivative in b brings down one power of x :

$$\frac{\partial^2}{\partial b^2} \int e^{-ax^2+bx} dx \Big|_{b=0} = \int x^2 e^{-ax^2} dx. \quad (0.2.29)$$

Do the left side from the closed form: $\partial_b^2 [\sqrt{\pi/a} e^{b^2/4a}] = \sqrt{\pi/a} e^{b^2/4a} [\frac{1}{2a} + \frac{b^2}{4a^2}]$, which at $b = 0$ is $\frac{1}{2a} \sqrt{\pi/a}$. Same answer as (0.2.27). ✓

Now read what just happened structurally. We built a function of a source, $Z(b)$; differentiating it with respect to the source and switching the source off produced an *average* of a power of the integration variable. Rename $Z(b)$ the generating functional, rename x the field, rename "average of a power" a correlation function, and you have the working method of quantum field theory, in which every Feynman diagram is a term produced by differentiating $e^{b^2/4a}$ enough times. The whole apparatus is (0.2.25) plus bookkeeping.

▼ Grind box — when you may differentiate under the integral sign, and when you may not

The swap in (0.2.27) is a genuine hypothesis, not a formality. It is a statement that a limit (the derivative in a) commutes with another limit (the integral), and limits do not generally commute.

A concrete failure. Let $h(x, a) = \frac{2x}{a^2} e^{-x^2/a^2}$ for $a > 0$. Substituting $u = x^2/a^2$,

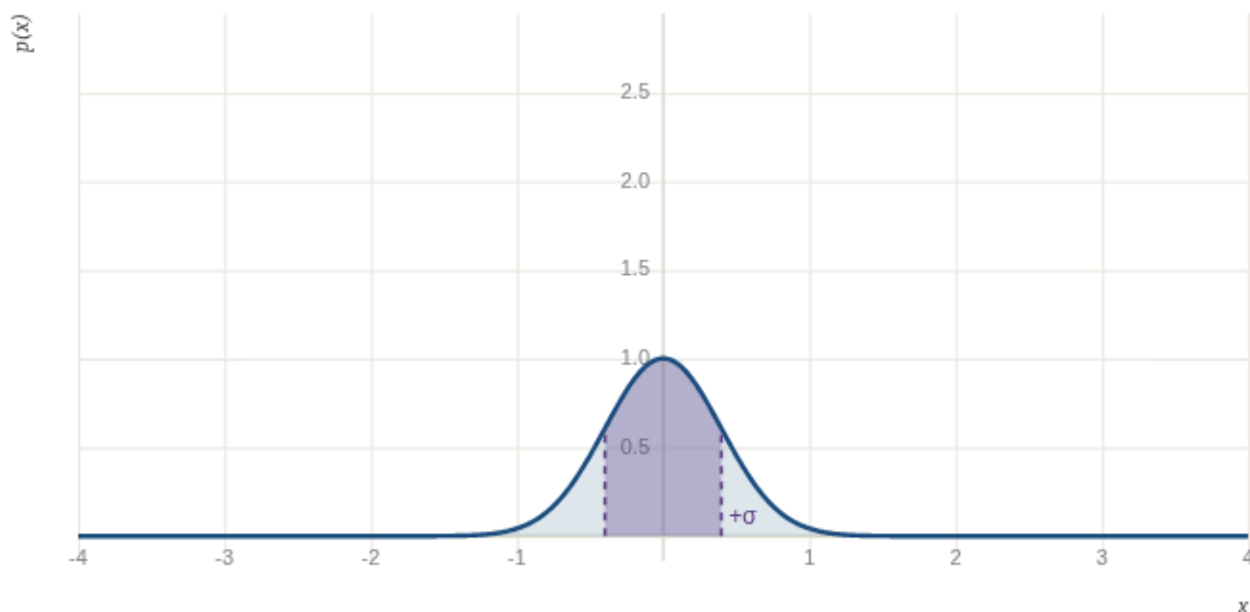
$$\int_0^1 h(x, a) dx = \int_0^{1/a^2} e^{-u} du = 1 - e^{-1/a^2} \xrightarrow{a \rightarrow 0} 1.$$

But for any fixed $x > 0$, $h(x, a) \rightarrow 0$ as $a \rightarrow 0$ (the exponential beats the prefactor). So $\lim \int = 1$ while $\int \lim = 0$. The mass has not vanished; it has escaped into an ever narrower, ever taller spike near the origin. Hold that image: it is the Dirac delta being born, and Chapter 0.9 makes it respectable.

The condition that saves us. The sufficient hypothesis (dominated convergence, proved properly in Chapter 4.3) is that the a -derivative of the integrand be bounded, uniformly in a near the point of interest, by a single fixed integrable function. Check it for our case: fix $a_0 > 0$ and let a range over $[a_0, \infty)$. Then

$$\left| \frac{\partial}{\partial a} e^{-ax^2} \right| = x^2 e^{-ax^2} \leq x^2 e^{-a_0 x^2},$$

and the right-hand side is a fixed function of x with finite integral (it is (0.2.27) itself, which we have just evaluated as finite). So the swap is justified for every $a > 0$, which is every case we care about. The moral is the book's general one: the manipulation is legal, and you should be able to say *why* on demand.



width σ $\sigma = 0.398$ peak = 1.0021 peak $\times \sigma = 0.39894$

total area = 1.00000 area in $\pm\sigma = 0.68269$

Area pinned, shape forced. The curve is the normalised Gaussian $p(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-x^2/2\sigma^2}$, whose total integral is 1 by (0.2.23) with $a = 1/2\sigma^2$. The readouts are computed numerically by Simpson's rule from §1, not from the formula, so the constancy of the total area is *measured*, not asserted. Slide σ and watch two invariants hold: the total area stays at 1.00000, and the shaded band between $\pm\sigma$ (dashed) always holds 0.68269 of it, whatever the width. The consequence is the one worth carrying: since the area cannot change, **narrowing the curve forces the peak up as $1/\sigma$** — the readout peak $\times \sigma$ is a constant $1/\sqrt{2\pi} = 0.3989$. You cannot have a distribution that is both narrow and short. Push σ to its minimum and you are watching the Dirac delta of Chapter 0.9 form; the same height-versus-width trade, applied to a wave function and its Fourier transform, is the uncertainty principle (Problem 4).

5 · Improper integrals, and when things diverge

Every integral in §4 ran to infinity, so we should be precise about what that means. An integral over an unbounded region is *defined* as a limit of integrals over bounded ones:

$$\int_1^\infty f(x) \, dx \equiv \lim_{R \rightarrow \infty} \int_1^R f(x) \, dx, \quad (0.2.30)$$

and it exists only if that limit does. The canonical test case is the power law, and it is worth doing once because it decides essentially every convergence question in physics.

For $p \neq 1$, (0.2.12) gives

$$\int_1^R x^{-p} dx = \left[\frac{x^{1-p}}{1-p} \right]_1^R = \frac{R^{1-p} - 1}{1-p}, \quad (0.2.31)$$

and everything hinges on the sign of $1 - p$. If $p > 1$ then $1 - p < 0$, so $R^{1-p} \rightarrow 0$ and the integral converges to $1/(p - 1)$. If $p < 1$ then $R^{1-p} \rightarrow \infty$ and it diverges. The borderline $p = 1$ has to be done separately because the antiderivative changes character: $\int_1^R x^{-1} dx = \ln R \rightarrow \infty$. So

$$\int_1^\infty \frac{dx}{x^p} \text{ converges } \iff p > 1, \quad \int_0^1 \frac{dx}{x^p} \text{ converges } \iff p < 1. \quad (0.2.32)$$

The second half of (0.2.32) is the identical computation with the limit taken at the other end ($\int_\epsilon^1 x^{-p} dx = (1 - \epsilon^{1-p})/(1 - p)$), and it says something you should hold onto: **the same integrand can fail at either end, and for opposite reasons.** $1/x$ diverges at both. Nothing converges at both ends by being a pure power law — you need different behaviour in the two regimes, which is why real physical answers always have a scale in them somewhere.

Which brings us to the payoff. The quantum corrections of Chapter 5.9 look like $\int^\Lambda d^4k/k^4 \sim \ln \Lambda$ and diverge as the cutoff is removed — which is not a new pathology of quantum field theory but exactly (0.2.32) in momentum space, failing at the large- k ("ultraviolet") end, with the small- k ("infrared") end able to fail independently for the other reason. **Renormalization** is the disciplined handling of that: such an integral was never a physical quantity on its own, and what you actually measure is a combination in which the divergence cancels and a finite, scale-dependent answer survives.

FAMILIAR GROUND

You have been computing definite integrals professionally for years. The area under the plasma concentration–time curve is one:

$$\text{AUC} = \int_0^{\infty} C(t) dt.$$

Take first-order elimination, $C(t) = C_0 e^{-kt}$. An antiderivative of e^{-kt} is $-e^{-kt}/k$, so by (0.2.12) and (0.2.30),

$$\text{AUC} = C_0 \lim_{R \rightarrow \infty} \left[\frac{-e^{-kt}}{k} \right]_0^R = C_0 \lim_{R \rightarrow \infty} \frac{1 - e^{-kR}}{k} = \frac{C_0}{k}.$$

With $C_0 = \text{Dose}/V$ this gives $\text{AUC} = \text{Dose}/(Vk)$, i.e. $CL = \text{Dose}/\text{AUC} = Vk$ — the identity you use without thinking, now derived in three lines from the definition of an integral.

But the structural point is better than the formula. Define cumulative exposure $E(t) = \int_0^t C$. Then (0.2.9) says $E'(t) = C(t)$: **concentration is the rate of accumulation of exposure**. AUC is not "a summary statistic that happens to be an area" — it is the accumulated quantity of which the measured concentration is precisely the derivative. That is why AUC, not $C_{\max}$, is the exposure metric that pairs with a clearance, and it is why Problem 3's infusion gives the same AUC as a bolus: the total accumulated is fixed by the total delivered and the rate constant, no matter how you schedule it.

Now the bridge, and it is a real one rather than a pun. In Chapter 1.2 the central object of all of classical and quantum physics is

$$S[\text{path}] = \int_{t_1}^{t_2} L(q(t), \dot{q}(t)) dt,$$

the *action*: a single number obtained by accumulating a rate (the Lagrangian, in units of energy) over an entire history. Structurally that is your AUC — one number summarising a whole time course. The difference is what you do with it. AUC is a number you measure for the history that happened. The action is a number you compute for *every history that might have happened*, and the law of physics is that nature picks the history where S is stationary. Chapter 1.2 makes that precise, and by then you will already know that the way you extract a law from an integral like that is integration by parts.

△ WHY THIS ISN'T OBVIOUS

Differentiation is an *algorithm*. Hand me any function built from powers, exponentials, logs and trig functions by adding, multiplying, dividing and composing, and the rules of Chapter 0.1 will grind out its derivative mechanically, and the answer will be a function of the same kind. The class is closed.

Integration is not an algorithm, and the class is not closed. There is no procedure that antidifferentiates an arbitrary elementary function, because *the antiderivative frequently is not an elementary function*. The examples are not obscure:

$$e^{-x^2}, \quad \frac{\sin x}{x}, \quad \frac{e^x}{x}, \quad \sqrt{1+x^4}, \quad \frac{1}{\ln x}$$

none of which has an elementary antiderivative. This is a **theorem** (Liouville, 1835; Risch later turned it into an actual decision procedure), not a confession that nobody has been clever enough. There is no answer to find.

Three consequences, all of which shape this book. First, §4 was not evasion — squaring and going to polar is the *only* way in, and note that it delivers the integral over the whole line and gives you nothing at all for $\int_0^1 e^{-x^2} dx$, which has no closed form and never will. Second, the handful of integrals that *can* be done — Gaussians, and things reducible to Gaussians — therefore carry a wildly disproportionate load in theoretical physics. Third, and least comfortable: when you notice that Gaussians are everywhere in physics, part of the reason is genuinely physical (the central limit theorem, Chapter 0.9; the fact that any smooth minimum looks quadratic, Chapter 0.3), and part of the reason is that Gaussians are what we can integrate. Perturbation theory in quantum field theory is, structurally, the practice of expanding around the one integral we can do. It is worth knowing which part of a formalism is nature and which part is our own limitations.

6 · Worked examples

WORKED EXAMPLE 1 — ESCAPE VELOCITY, FROM AN INTEGRAL OVER AN INFINITE REGION

How fast must you throw something so that it never comes back?

Newton's gravitational force on a mass m at distance r from a body of mass M has magnitude GMm/r^2 , directed inward. To move the mass from the surface $r = R$ out to $r = \infty$ against that force, you must supply work equal to the accumulated force-times-distance — which is exactly a definite integral, because the force is not constant along the way:

$$W = \int_R^\infty \frac{GMm}{r^2} dr = GMm \lim_{\rho \rightarrow \infty} \left[-\frac{1}{r} \right]_R^\rho = GMm \lim_{\rho \rightarrow \infty} \left(\frac{1}{R} - \frac{1}{\rho} \right) = \frac{GMm}{R}.$$

The integral converges because the exponent is $p = 2 > 1$, exactly the criterion of (0.2.32). This is not a technicality: it is the reason escape is possible at all. An infinite journey costs a finite amount of energy *only because gravity weakens faster than $1/r$* . If gravity fell off as $1/r$ the integral would be ln-divergent and nothing could ever escape anything.

Now supply that energy from the kinetic energy the object starts with. Setting $\frac{1}{2}mv_{\text{esc}}^2 = GMm/R$ and solving,

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}}.$$

For Earth, $GM = 3.986 \times 10^{14} \text{ m}^3\text{s}^{-2}$ and $R = 6.371 \times 10^6 \text{ m}$, giving $v_{\text{esc}} = 1.119 \times 10^4 \text{ m s}^{-1}$, about 11.2 km s^{-1} . The mass m cancelled, which is the equivalence principle showing up uninvited three parts early (Chapter 3.1).

One more thing to notice, because it is the Fundamental Theorem leaving a fingerprint on physical convention. The antiderivative $-GMm/r$ is only defined up to a constant, and we chose the constant by taking the reference point at infinity. That choice is why gravitational potential energy is *negative* for bound objects. The constant is physically meaningless — only differences of potential energy appear in (0.2.12) — but the convention matters for reading signs off the page. In general relativity the same freedom is far less innocent: there, energy itself has no local definition, for reasons traceable to this ambiguity (Chapter 3.5).

WORKED EXAMPLE 2 — THE WIDTH OF A GAUSSIAN, BY PARAMETER DIFFERENTIATION

For the normalised distribution $\rho(x) = \mathcal{N}e^{-ax^2}$, find $\langle x^2 \rangle$.

First the normalisation. Demanding $\int \rho = 1$ and using (0.2.23):

$$\mathcal{N}\sqrt{\frac{\pi}{a}} = 1 \quad \implies \quad \mathcal{N} = \sqrt{\frac{a}{\pi}}.$$

Now the second moment. By definition $\langle x^2 \rangle = \int x^2 \rho(x) dx$, and rather than attack that integral we use the parameter trick of §4.4. Write the normalisation identity with the parameter left visible,

$$Z(a) \equiv \int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\pi} a^{-1/2},$$

and differentiate both sides with respect to a . On the left, ∂_a acting on the integrand brings down $-x^2$; on the right it is one line of power rule:

$$-\int x^2 e^{-ax^2} dx = \frac{dZ}{da} = -\frac{1}{2}\sqrt{\pi} a^{-3/2}.$$

Therefore $\int x^2 e^{-ax^2} dx = \frac{1}{2}\sqrt{\pi} a^{-3/2}$, and dividing by $Z(a) = \sqrt{\pi} a^{-1/2}$ to normalise:

$$\boxed{\langle x^2 \rangle = \frac{\frac{1}{2}\sqrt{\pi} a^{-3/2}}{\sqrt{\pi} a^{-1/2}} = \frac{1}{2a}.$$

Sanity check against the form you know: the statistician's Gaussian is $e^{-x^2/2\sigma^2}$, i.e. $a = 1/2\sigma^2$, and the formula returns $\langle x^2 \rangle = \sigma^2$. ✓ The variance is the reciprocal of twice the exponent's coefficient, which is the sanest way to read a Gaussian: **a big a means a sharply peaked, narrow distribution.**

Observe the economy. We never integrated $x^2 e^{-ax^2}$. We differentiated a *known answer* with respect to a parameter that was sitting there anyway. In Chapter 5.5 this is done with a source rather than a coefficient (§4.5), the integration variable is a field, and the resulting " $\langle x^2 \rangle$ " is the propagator — but it is this calculation.

And it is about to be a physical prediction. In Chapter 4.4 a free particle in a Gaussian wave packet has $\psi(x) \propto e^{-ax^2}$, so its probability density is $|\psi|^2 \propto e^{-2ax^2}$ — the same shape with $a \rightarrow 2a$. The position uncertainty is therefore

$$(\Delta x)^2 = \langle x^2 \rangle = \frac{1}{2(2a)} = \frac{1}{4a}, \quad \Delta x = \frac{1}{2\sqrt{a}}.$$

Problem 4 computes the width of the same packet in momentum space and finds $\Delta k = \sqrt{a}$. Multiply: $\Delta x \Delta k = \frac{1}{2}$, independent of a . That is the Heisenberg uncertainty relation, exactly saturated, and you will have derived it in Chapter 0.2 using nothing but a Gaussian integral and a derivative with respect to a parameter.

7 · Your turn

PROBLEM 1 — BY PARTS TWICE, THEN ALL AT ONCE

Evaluate $\int_0^\infty x^2 e^{-x} dx$ by integrating by parts twice, checking the boundary terms rather than assuming them. Then define $I_n = \int_0^\infty x^n e^{-x} dx$ and derive a reduction formula relating I_n to I_{n-1} ; solve it. Finally, write down the indefinite integral $\int x^2 e^{-x} dx$ and verify it by differentiating.

– Solution

The definite integral. Take $u = x^2$, $v' = e^{-x}$, so $u' = 2x$ and $v = -e^{-x}$. Then (0.2.16) gives

$$\int_0^\infty x^2 e^{-x} dx = \left[-x^2 e^{-x} \right]_0^\infty + 2 \int_0^\infty x e^{-x} dx.$$

The boundary term: at $x = 0$ the factor x^2 kills it; as $x \rightarrow \infty$, e^{-x} decays faster than any power grows, so $x^2 e^{-x} \rightarrow 0$. (That claim is worth one line: write $e^x > x^3/6$ for $x > 0$, which follows from the series, so $x^2 e^{-x} < 6/x \rightarrow 0$.) The boundary term is genuinely zero, checked.

Repeat on the survivor with $u = x$, $v' = e^{-x}$:

$$\int_0^\infty x e^{-x} dx = \left[-x e^{-x} \right]_0^\infty + \int_0^\infty e^{-x} dx = 0 + \left[-e^{-x} \right]_0^\infty = 1.$$

Hence $\int_0^\infty x^2 e^{-x} dx = 2 \cdot 1 = 2$.

Reduction formula. The same step with $u = x^n$ gives, for $n \geq 1$,

$$I_n = \left[-x^n e^{-x} \right]_0^\infty + n \int_0^\infty x^{n-1} e^{-x} dx = n I_{n-1}.$$

With $I_0 = \int_0^\infty e^{-x} dx = 1$, iterating gives $I_n = n!$. Check: $I_2 = 2$. ✓

This function of n is the gamma function, $\Gamma(n+1) = n!$, and the reduction formula you just derived is the reason Γ interpolates the factorials to non-integer arguments — which is how dimensional regularization in Chapter 5.9 gets away with computing integrals in $4 - \epsilon$ dimensions.

Indefinite. Running by parts without limits (or just guessing a quadratic times e^{-x} and matching):

$$\int x^2 e^{-x} dx = -(x^2 + 2x + 2)e^{-x} + C.$$

Check by the product rule: $\frac{d}{dx} [-(x^2 + 2x + 2)e^{-x}] = -(2x + 2)e^{-x} + (x^2 + 2x + 2)e^{-x} = x^2 e^{-x}$. ✓
Evaluating from 0 to ∞ : $0 - (-2) = 2$, agreeing with the definite calculation.

PROBLEM 2 — PARAMETER DIFFERENTIATION, IN ITS PUREST FORM

Show that $\int_0^\infty e^{-ax} dx = 1/a$ for $a > 0$. Then, by differentiating n times with respect to a , obtain $\int_0^\infty x^n e^{-ax} dx = n!/a^{n+1}$ without integrating by parts even once. Check it against Problem 1.

– Solution

The base integral. An antiderivative of e^{-ax} is $-e^{-ax}/a$, so by (0.2.30) and (0.2.12),

$$\int_0^\infty e^{-ax} dx = \lim_{R \rightarrow \infty} \frac{1 - e^{-aR}}{a} = \frac{1}{a} \quad (a > 0).$$

The condition $a > 0$ is where convergence lives; for $a \leq 0$ the integrand does not decay and there is nothing to compute.

Differentiate. Each ∂_a on the integrand brings down a factor $-x$:

$$\frac{d^n}{da^n} e^{-ax} = (-x)^n e^{-ax}.$$

On the right-hand side, differentiate a^{-1} repeatedly: $\frac{d}{da} a^{-1} = -a^{-2}$, then $+2a^{-3}$, then $-6a^{-4}$, and in general $\frac{d^n}{da^n} a^{-1} = (-1)^n n! a^{-(n+1)}$ — provable by induction in one line, since differentiating $(-1)^n n! a^{-(n+1)}$ gives $(-1)^{n+1} (n+1)! a^{-(n+2)}$. Equating the two sides,

$$(-1)^n \int_0^\infty x^n e^{-ax} dx = (-1)^n \frac{n!}{a^{n+1}} \quad \implies \quad \int_0^\infty x^n e^{-ax} dx = \frac{n!}{a^{n+1}}.$$

Check. Setting $a = 1$ recovers $I_n = n!$ from Problem 1, which was obtained by n applications of integration by parts. Here it took one integral and a pattern in derivatives.

Legality. As in the grind box of §4.4, restrict a to $[a_0, \infty)$ with $a_0 > 0$; then $|x^n e^{-ax}| \leq x^n e^{-a_0 x}$, a fixed integrable function, so the swap of ∂_a and $\int$ is justified. Note this is also a compact proof that $\Gamma(n+1) = n!$, and that the whole family of "∫ polynomial times exponential" integrals is generated by one seed and the power rule. That is the pattern to steal: *find the seed integral with a parameter in it, then differentiate.*

PROBLEM 3 — AUC DOES NOT CARE ABOUT THE SCHEDULE

A drug is given by constant-rate IV infusion at rate R (mass per time) for a duration T , then stopped. Elimination is first order with rate constant k from a single compartment of volume V , so the concentration obeys $V \frac{dC}{dt} = R\Theta(t) - kVC$, where Θ is 1 during the infusion and 0 after.

(a) Solve for $C(t)$ during and after the infusion. (b) Compute $\text{AUC} = \int_0^\infty C \, dt$ and show it equals $\text{Dose}/(Vk)$, identical to a bolus of the same total dose. (c) Then obtain the same result in two lines without solving the differential equation at all.

– Solution

(a) During infusion, $\frac{dC}{dt} = R/V - kC$. The steady state is $C_\infty = R/(kV)$; writing $C = C_\infty + u$ turns the equation into $\dot{u} = -ku$, whose solution (Chapter 0.1, §5) is $u = u_0 e^{-kt}$. With $C(0) = 0$ we need $u_0 = -C_\infty$, so for $0 \leq t \leq T$

$$C(t) = \frac{R}{kV} (1 - e^{-kt}).$$

After the infusion stops, elimination is pure first order from $C_T \equiv C(T)$:

$$C(t) = C_T e^{-k(t-T)}, \quad C_T = \frac{R}{kV} (1 - e^{-kT}).$$

(b) Split the integral at T . The tail is the Familiar-ground calculation: $\int_T^\infty C = C_T/k$. The infusion phase:

$$\int_0^T \frac{R}{kV} (1 - e^{-kt}) \, dt = \frac{R}{kV} \left[t + \frac{e^{-kt}}{k} \right]_0^T = \frac{R}{kV} \left(T - \frac{1 - e^{-kT}}{k} \right).$$

Add them, writing $C_T/k = \frac{R}{k^2 V} (1 - e^{-kT})$:

$$\text{AUC} = \frac{RT}{kV} - \frac{R(1 - e^{-kT})}{k^2 V} + \frac{R(1 - e^{-kT})}{k^2 V} = \frac{RT}{kV} = \frac{\text{Dose}}{Vk} = \frac{\text{Dose}}{CL},$$

since the total dose delivered is RT . Every trace of T beyond the total dose cancelled exactly. Infuse it over a minute or over a week: same AUC.

(c) **The two-line version, which is the point of the problem.** Do not solve anything. Integrate the differential equation itself from 0 to ∞ , using (0.2.12) on the left:

$$\begin{aligned} V \int_0^\infty \frac{dC}{dt} \, dt &= \int_0^\infty R \Theta(t) \, dt - kV \int_0^\infty C \, dt \\ \implies V [C(\infty) - C(0)] &= \text{Dose} - kV \cdot \text{AUC}. \end{aligned}$$

With $C(0) = C(\infty) = 0$ the left side vanishes, and

$$\text{AUC} = \frac{\text{Dose}}{kV} = \frac{\text{Dose}}{CL}.$$

No solution of the ODE, no case-splitting, no exponentials. This is the Fundamental Theorem doing exactly what the insight callout in §2 advertised: a *local* statement (the mass-balance differential equation, true at each instant) has been converted into a *global* conservation statement (everything that went in eventually came out) by integrating, with only boundary values surviving. It also shows that $\text{AUC} = \text{Dose}/CL$ holds for *any* input schedule whatsoever — oral, infused, divided, erratic — provided elimination is linear. That robustness is not a pharmacological coincidence; it is a boundary-term argument.

PROBLEM 4 — THE FOURIER TRANSFORM OF A GAUSSIAN

Show that for $a > 0$

$$\int_{-\infty}^{\infty} e^{-ax^2} \cos(bx) \, dx = \sqrt{\frac{\pi}{a}} e^{-b^2/4a}.$$

Do it by parameter differentiation in b : differentiate under the integral sign, integrate the result by parts, and notice that you have obtained a differential equation for the answer. Then say what the result means about widths.

– Solution

Define $J(b) = \int_{-\infty}^{\infty} e^{-ax^2} \cos(bx) \, dx$. From (0.2.23), $J(0) = \sqrt{\pi/a}$.

Differentiate in b . The integrand's b -derivative is $-x e^{-ax^2} \sin(bx)$, dominated in absolute value by $|x|e^{-ax^2}$, which is integrable and independent of b — so the swap is legal:

$$J'(b) = - \int_{-\infty}^{\infty} x e^{-ax^2} \sin(bx) \, dx.$$

Integrate by parts. The factor $x e^{-ax^2}$ is a derivative in disguise: $\frac{d}{dx} \left(-\frac{1}{2a} e^{-ax^2} \right) = x e^{-ax^2}$. So take $v' = x e^{-ax^2}$, $v = -\frac{1}{2a} e^{-ax^2}$ and $u = \sin(bx)$, $u' = b \cos(bx)$ in (0.2.16):

$$J'(b) = - \left\{ \left[-\frac{\sin(bx)}{2a} e^{-ax^2} \right]_{-\infty}^{\infty} + \frac{b}{2a} \int_{-\infty}^{\infty} e^{-ax^2} \cos(bx) \, dx \right\}.$$

The boundary term vanishes because $e^{-ax^2} \rightarrow 0$ at both ends while $\sin$ stays bounded — checked, not assumed. What survives is

$$J'(b) = -\frac{b}{2a} J(b).$$

Solve. This is a first-order linear equation of exactly the kind in Chapter 0.1 §5, except that the rate is proportional to b rather than constant. Separating, $\frac{d}{db} \ln J = -b/2a$, so $\ln J = -b^2/4a + \text{const}$, i.e. $J(b) = J(0) e^{-b^2/4a}$. With $J(0) = \sqrt{\pi/a}$:

$$\int_{-\infty}^{\infty} e^{-ax^2} \cos(bx) \, dx = \sqrt{\frac{\pi}{a}} e^{-b^2/4a}. \quad \checkmark$$

(Cross-check: since $\int e^{-ax^2} \sin(bx) \, dx = 0$ by oddness, this is the real and complete content of $\int e^{-ax^2} e^{ibx} \, dx$ — which is (0.2.25) with $b \rightarrow ib$. The two agree, which is the first hint that (0.2.25) continues to hold for complex b ; Chapter 0.4 makes that legitimate.)

What it means. This is the *Fourier transform of a Gaussian*, and it is a *Gaussian* — the only function shape that is its own transform. Look at the widths. Reading off Worked Example 2, a factor e^{-ax^2} corresponds to a probability density e^{-2ax^2} with $\Delta x = 1/(2\sqrt{a})$. The transform $e^{-b^2/4a}$ corresponds to a density $e^{-b^2/2a}$ with $\Delta b = \sqrt{a}$. Their product is

$$\Delta x \Delta b = \frac{1}{2\sqrt{a}} \cdot \sqrt{a} = \frac{1}{2},$$

independent of a . Squeeze the function in x and its transform spreads in b by exactly the reciprocal amount — which is what the width slider in §4's figure was showing you geometrically for a single Gaussian, now made into a statement relating two functions.

Chapter 0.9 proves the general inequality $\Delta x \Delta b \geq \frac{1}{2}$ for *any* function, with equality only for Gaussians — the bandwidth theorem, which is a fact about waves and has nothing to do with quantum mechanics. Quantum mechanics then contributes one physical identification, $b = p/\hbar$ (Chapter 4.4), and the same line reads $\Delta x \Delta p \geq \hbar/2$. You have just done most of the work for the Heisenberg uncertainty principle, in a chapter about integration, using nothing but by parts and a derivative with respect to a parameter.

THE BRICK YOU JUST LAID

You have the integral as a **limit of tagged sums** ((0.2.2)), with an honest account of what has to be true for that limit to exist and a preview of why the definition gets replaced. You have both halves of the **Fundamental Theorem** derived from Chapter 0.1's linearization — the first by observing that a thin sliver is a rectangle up to second order, the second by noticing that telescoped interior terms cancel and only the boundary survives. You have **substitution** and **integration by parts** as the chain and product rules run backwards, and you know that "the boundary term vanishes" is a physical claim. You can do the **Gaussian integral** in all three forms, and you can generate its moments by **differentiating with respect to a parameter**. And you know exactly when an improper integral converges, which is the same question as when a quantum field theory diverges.

Where this gets spent.

- **The Fundamental Theorem** — Chapter 0.7, where interior-cancellation becomes Green's, Stokes' and the divergence theorems; Chapter 3.5, where local conservation laws are integrated into global ones (and where the attempt sometimes fails, informatively).
- **Integration by parts** — Chapter 1.2, where it converts stationary action into the Euler–Lagrange equation; Chapter 5.2, where the identical step produces every classical field equation; Chapter 4.3, where it makes $-i\hbar\partial_x$ Hermitian and thereby makes momentum an observable.
- **The Gaussian integral** — Chapter 4.4 (normalising a wave packet), Chapter 5.8 (the free propagator, where the Gaussian is oscillatory and you meet the $i\epsilon$ prescription), Chapter 5.5 (where all of perturbative QFT is (0.2.25) plus corrections).
- **Parameter differentiation** — Chapter 5.5, where differentiating with respect to a source is how every correlation function and every Feynman diagram is extracted from a generating functional.
- **Convergence of improper integrals** — Chapter 5.9, where the failure of (0.2.32) at large momentum is the ultraviolet divergence and renormalization is what you do about it.

Next, Chapter 0.3 takes the observation that kept surfacing here — that the leftover error after linearizing is quadratic, and after that cubic, and so on — and turns it into a systematic expansion. That is the tool physicists reach for more often than any other, because almost nothing is solvable and almost everything is *nearly* solvable.